

NOTE 01 — WHAT IS A SOC?

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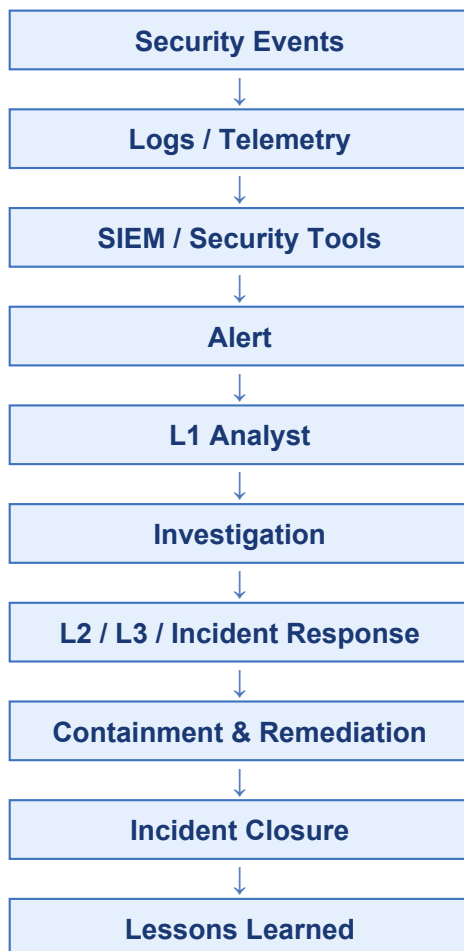
A Security Operations Center (SOC) is a centralized security function responsible for monitoring, detecting, investigating, and responding to cybersecurity threats affecting an organization's systems, networks, applications, endpoints, and data.

The SOC acts as an organization's security monitoring and response center.

Main objectives of a SOC

- Detect suspicious activity.
- Monitor security alerts.
- Investigate potential incidents.
- Identify compromised systems.
- Respond to security incidents.
- Reduce the impact of attacks.
- Collect and analyze security logs.
- Escalate serious incidents.
- Improve security controls based on lessons learned.

Simple SOC workflow



What does a SOC monitor?

Area	Examples
Endpoints	Windows/Linux computers
Network	Firewalls, routers, IDS/IPS
Applications	Web applications, APIs
Identity	Login and authentication events
Cloud	Cloud infrastructure and services
Email	Phishing and malicious attachments
Servers	Windows/Linux servers
Security tools	SIEM, EDR, firewall, antivirus
Threat intelligence	Malicious IPs, domains, hashes

Common SOC technologies

SIEM

Security Information and Event Management

A SIEM collects and correlate security logs from different sources.

Examples:

- Splunk
- Microsoft Sentinel
- Elastic Security
- IBM QRadar

EDR

Endpoint Detection and Response

EDR monitors endpoint activity and helps detect suspicious processes, files, persistence, and other endpoint threats.

SOAR

Security Orchestration, Automation and Response

SOAR platforms automate repetitive security tasks and response workflows.

Threat Intelligence

Threat intelligence provides information about known or suspected threats such as:

- Malicious IP addresses
- Malicious domains
- File hashes
- Threat actors
- Malware families
- Attack techniques

LetsDefend's SOC Fundamentals course specifically covers SOC types, roles, SIEM, log management, EDR, SOAR, threat intelligence, and common analyst mistakes. LetsDefend

What does a SOC analyst do?

A SOC analyst generally:

1. Monitors security alerts.
2. Performs initial alert triage.
3. Investigates suspicious activity.
4. Determines whether an alert is a false positive or potential threat.
5. Collects evidence.
6. Escalates serious incidents.
7. Documents investigations.
8. Helps contain and remediate incidents.

Conclusion

A SOC provides continuous security monitoring and helps an organization identify and respond to cyber threats. It combines people, processes, and technology to improve an organization's defensive security capability.